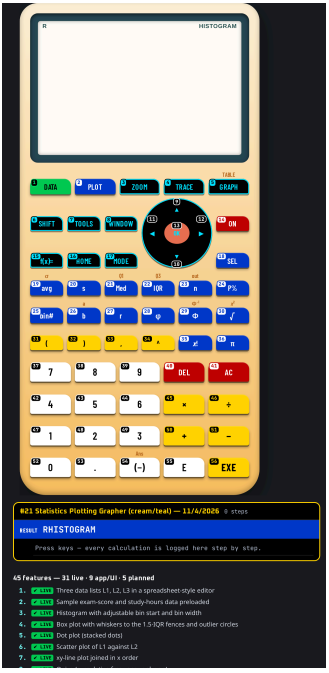
U-45 CALCULATORS

User Manual

Calculator #21 Statistics Plotting Grapher (cream/teal)

Graphing – colour



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#21 — Statistics Plotting Grapher (cream/teal)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 DATA]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

statistics plotting from lists L1–L3 (histogram, box, scatter, ogive, normal curve).

Chapters

1. Getting started
2. Features
3. Feature guide (2b)
4. Numbered key map
5. Key-by-key reference
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8. Keyboard shortcuts
9. Common mistakes
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1 · Getting started

Open U-45 CALCULATORS, choose **#21** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#37, #38, #39, #42, #43, #44, #47, #48, #49, #52, #53, #54, #55
op	Operators & execute	#13, #31, #32, #33, #34, #45, #46, #50, #51, #56
fn	Functions	#2, #18, #19, #20, #21, #22, #23, #24, #25, #26, #27, #28, #29, #30, #35, #36
mem	Memory, variables & registers	#1
clr	Clear, delete & power	#14, #40, #41

mod	**Shift, modes, menus & navigation**	**#3, #4, #5, #6, #7, #8, #9, #10, #11, #12, #15, #16, #17**

2 · Features

45 features: 31 live on the keys, 9 app/UI features, 5 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **Three data lists L1, L2, L3 in a spreadsheet-style editor**
2. LIVE **Sample exam-score and study-hours data preloaded**
3. LIVE **Histogram with adjustable bin start and bin width**
4. LIVE **Box plot with whiskers to the 1.5·IQR fences and outlier circles**
5. LIVE **Dot plot (stacked dots)**
6. LIVE **Scatter plot of L1 against L2**
7. LIVE **xy-line plot joined in x order**
8. LIVE **Ogive (cumulative frequency polygon)**
9. LIVE **Normal probability plot with reference line**
10. LIVE **Pie chart of bin frequencies with labels**
11. LIVE **Normal curve fitted to L1 with ±1σ, ±2σ guides**
12. LIVE **Trace bins with count, cumulative and relative frequency**
13. LIVE **Trace box-plot five numbers**
14. LIVE **Trace scatter points in order**
15. LIVE **Trace the normal curve with z, pdf and P(X≤x)**
16. LIVE **ZoomStat automatic window for every plot type**
17. LIVE **Fit overlay on scatter: linear, quadratic or exponential with R^2**
18. LIVE **Shade a normal area between two values with probability**
19. LIVE **Five-number summary screen**
20. LIVE **Outlier report with fences**

21. LIVE Stem-and-leaf display
22. LIVE Frequency table with cf and %
23. LIVE Percentile with linear interpolation
24. LIVE Sort L1, load sample data, clear lists
25. LIVE Optional overlay curve y = f(x) on any plot
26. LIVE PLOT key cycles the nine plot types
27. LIVE Home list statistics avg(), sdev(), psd(), cnt(), med(), q1(), q3(), iqr(), outl()
28. LIVE Home pctl(k,p) and bincount(i)
29. LIVE Home regression slope(), icept(), corrl() for L1 vs L2
30. LIVE Home normal functions npdf(), ncdf(), invn()
31. LIVE Table view listing L1–L3 side by side
32. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
33. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
34. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
35. APP/UI Show / hide key numbers with one button (remembered between sessions)
36. APP/UI Copy the current display text to the clipboard
37. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
38. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
39. APP/UI Reference photo viewer and tool-tip name on every key

40. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
41. PLANNED **Graph image export (PNG)**
42. PLANNED **Shareable function list link**
43. PLANNED **Dark screen theme**
44. PLANNED **Speech read-out of values**
45. PLANNED **Exam mode**

2b · Feature guide

Calculator #21 is a statistics plotting from lists L1–L3 (histogram, box, scatter, ogive, normal curve). Each card below is one of its 31 working features, with the keys that reach it and, where one exists, a verified example.

F1 Three data lists L1, L2, L3 in a spreadsheet-style editor

Where: #1 DATA **Open or close the list editor**

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F2 Sample exam-score and study-hours data preloaded

Where: #1 DATA **Open or close the list editor;** #20 s **Sample s.d. of list k**

See it in action: Mean of the sample exam scores L1 — *avg(1)*= 75.55

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F3 Histogram with adjustable bin start and bin width

Where: #25 bin# Count in histogram bin i

See it in action: Histogram bin 3 count (bins of width 10 from 50) — *bincount(3)*= 8

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F4 Box plot with whiskers to the 1.5·IQR fences and outlier circles

Where: #2 PLOT Cycle plot type; #7 TOOLS Plot types, fits, summaries; #22 IQR Interquartile range

See it in action: Interquartile range of L1 (median-of-halves) — *iqr(1)*= 13.5

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F5 Dot plot (stacked dots)

Where: #2 PLOT Cycle plot type; #7 TOOLS Plot types, fits, summaries

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F6 Scatter plot of L1 against L2

Where: #2 PLOT Cycle plot type; #7 TOOLS Plot types, fits, summaries

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F7 xy-line plot joined in x order

Where: #2 PLOT Cycle plot type; #7 TOOLS Plot types, fits, summaries; #9 ▲ Cursor ▲ — up / previous history line / previous item

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F8 Ogive (cumulative frequency polygon)

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F9 Normal probability plot with reference line

Where: #29 Φ Normal probability a…b; #2 PLOT Cycle plot type; #7 TOOLS Plot types, fits, summaries

See it in action: Standard normal probability within one σ — *ncdf(−1,1)*= 0.68268949

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F10 Pie chart of bin frequencies with labels

Where: #25 bin# Count in histogram bin i

See it in action: Histogram bin 3 count (bins of width 10 from 50) — *bincount(3)*= 8

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F11 Normal curve fitted to L1 with ±1σ, ±2σ guides

Where: #15 f(x)= Overlay curve; #28 φ Normal density; #29 Φ Normal probability a…b

See it in action: Standard normal probability within one σ — *ncdf(−1,1)*= 0.68268949

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F12 Trace bins with count, cumulative and relative frequency

Where: #4 TRACE TRACE — move along a graph; #8 WINDOW Bins and view; #23 n Count of list k

See it in action: Histogram bin 3 count (bins of width 10 from 50) — *bincount(3)*= 8

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F13 Trace box-plot five numbers

Where: #2 PLOT Cycle plot type; #4 TRACE TRACE — move along a graph; #7 TOOLS Plot types, fits, summaries

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F14 Trace scatter points in order

Where: #4 TRACE TRACE — move along a graph

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F15 Trace the normal curve with z, pdf and P(X≤x)

Where: #4 TRACE TRACE — move along a graph; #15 f(x)= Overlay curve; #28 φ Normal density

See it in action: Standard normal probability within one σ — *ncdf(−1,1)*= 0.68268949

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F16 ZoomStat automatic window for every plot type

Where: #2 PLOT Cycle plot type; #3 ZOOM ZoomStat and zoom menu; #7 TOOLS Plot types, fits, summaries

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F17 Fit overlay on scatter: linear, quadratic or exponential with R^2

Where: #15 f(x)= Overlay curve

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F18 Shade a normal area between two values with probability

Where: #29 Φ Normal probability a…b; #28 φ Normal density; #33 , Comma / argument separator (between function arguments, list items or coordinates)

See it in action: Standard normal probability within one σ — *ncdf(−1,1)*= 0.68268949

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F19 Five-number summary screen

Where: #16 HOME HOME — return to the main calculation screen; #17 MODE SETUP — angle unit, number format, display options; #23 n Count of list k

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F20 Outlier report with fences

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F21 Stem-and-leaf display

Where: #17 MODE SETUP — angle unit, number format, display options

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F22 Frequency table with cf and %

Where: #5 GRAPH **GRAPH — draw the graph**

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F23 Percentile with linear interpolation

Where: #24 P% **Percentile**

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F24 Sort L1, load sample data, clear lists

Where: #1 DATA **Open or close the list editor;** #20 s **Sample s.d. of list k;** #41 AC **AC — clear all of the current calculation**

See it in action: Mean of the sample exam scores L1 — *avg(1)*= 75.55

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F25 Optional overlay curve y = f(x) on any plot

Where: #15 f(x)= Overlay curve; #2 PLOT Cycle plot type; #7 TOOLS Plot types, fits, summaries

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F26 PLOT key cycles the nine plot types

Where: #7 TOOLS Plot types, fits, summaries; #2 PLOT Cycle plot type

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F27 Home list statistics avg(), sdev(), psd(), cnt(), med(), q1(), q3(), iqr(), outl()

Where: #19 avg Mean of list k; #21 Med Median of list k; #1 DATA Open or close the list editor

See it in action: Mean of the sample exam scores L1 — *avg(1)*= 75.55; Median of L1 — *med(1)*= 75

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F28 Home pctl(k,p) and bincount(i)

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

See it in action: Histogram bin 3 count (bins of width 10 from 50) — *bincount(3)*= 8

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F29 Home regression slope(), icept(), corrl() for L1 vs L2

Where: #26 b Regression slope

See it in action: Least-squares slope of L1 on L2 — *slope()*= 4.15344759; Correlation coefficient r of L1 and L2 — *corrl()*= 0.98980838

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F30 Home normal functions npdf(), ncdf(), invn()

Where: #28 φ Normal density; #29 Φ Normal probability a...b

See it in action: Standard normal probability within one σ — *ncdf(−1,1)*= 0.68268949

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F31 Table view listing L1–L3 side by side

Where: #5 GRAPH GRAPH — draw the graph; #8 WINDOW Bins and view

Try it: Press Reset, use the key(s) above, and write the display here:

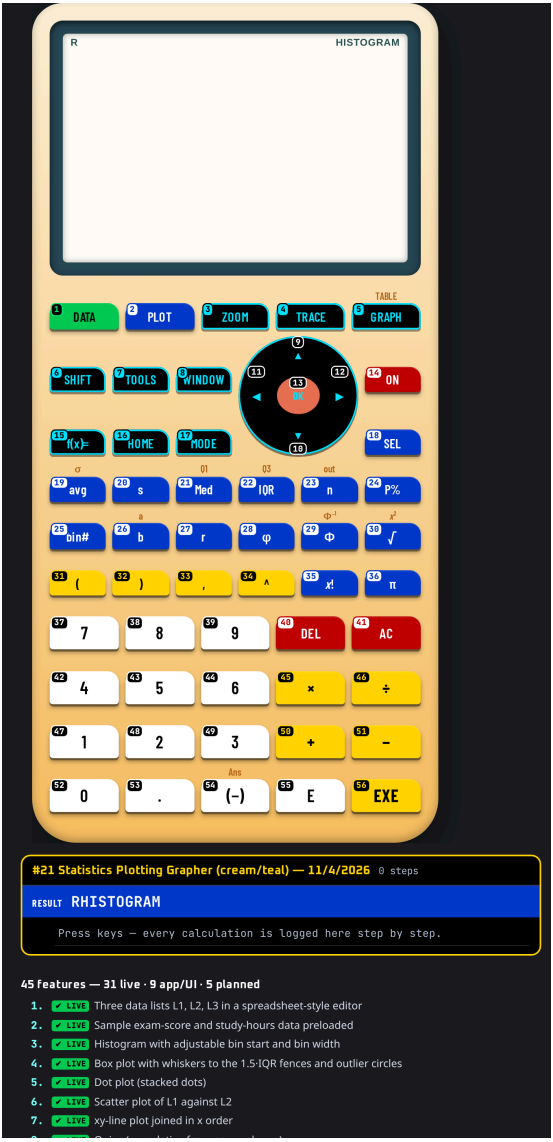
_____

Your notes:

_____

_____

3 · Numbered key map



Calculator #21 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	DATA	·	·	·	mem
2	PLOT	·	·	·	fn
3	ZOOM	·	·	·	mod
4	TRACE	·	·	·	mod
5	GRAPH	TABLE	·	·	mod
6	SHIFT	·	·	·	mod
7	TOOLS	·	·	·	mod
8	WINDOW	·	·	·	mod
9	▲	·	·	·	mod
10	▼	·	·	·	mod
11	◀	·	·	·	mod
12	▶	·	·	·	mod
13	OK	·	·	·	op
14	ON	·	·	·	clr
15	f(x)=	·	·	·	mod
16	HOME	·	·	·	mod
17	MODE	·	·	·	mod
18	SEL	·	·	·	fn
19	avg	σ	·	·	fn
20	s	·	·	·	fn
21	Med	Q1	·	·	fn
22	IQR	Q3	·	·	fn
23	n	out	·	·	fn

#	Face	Shift	Alpha	3rd	Group
24	P%	·	·	·	fn
25	bin#	·	·	·	fn
26	b	a	·	·	fn
27	r	·	·	·	fn
28	φ	·	·	·	fn
29	Φ	Φ^{-1}	·	·	fn
30	√	x^2	·	·	fn
31	(	·	·	·	op
32	)	·	·	·	op
33	,	·	·	·	op
34	^	·	·	·	op
35	x!	·	·	·	fn
36	π	·	·	·	fn
37	7	·	·	·	num
38	8	·	·	·	num
39	9	·	·	·	num
40	DEL	·	·	·	clr
41	AC	·	·	·	clr
42	4	·	·	·	num
43	5	·	·	·	num
44	6	·	·	·	num
45	×	·	·	·	op
46	÷	·	·	·	op
47	1	·	·	·	num
48	2	·	·	·	num
49	3	·	·	·	num
50	+	·	·	·	op
51	−	·	·	·	op

#	Face	Shift	Alpha	3rd	Group
52	0	·	·	·	num
53	.	·	·	·	num
54	(–)	Ans	·	·	num
55	E	·	·	·	num
56	EXE	·	·	·	op

4 · Key-by-key reference

Every key on calculator #21, in number order, with every function it carries in this app.

#1 DATA `Memory, variables & registers`

Primary: Open or close the list editor

#2 PLOT `Functions`

Primary: Cycle plot type

#3 ZOOM `Shift, modes, menus & navigation`

Primary: ZoomStat and zoom menu

#4 TRACE `Shift, modes, menus & navigation`

Primary: TRACE — move along a graph

#5 GRAPH `Shift, modes, menus & navigation`

Primary: GRAPH — draw the graph

shifted (2nd / f / SHIFT layer) (press #6 first): TABLE — TABLE — table of values

#6 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#7 TOOLS `Shift, modes, menus & navigation`

Primary: Plot types, fits, summaries

#8 WINDOW `Shift, modes, menus & navigation`

Primary: Bins and view

#9 ▲ `Shift, modes, menus & navigation`

Primary: Cursor ▲ — up / previous history line / previous item

#10 ▼ `Shift, modes, menus & navigation`

Primary: Cursor ▼ — down / next history line / next item

#11 ◀ `Shift, modes, menus & navigation`

Primary: Cursor ◀ — move left in the line

#12 ▶ `Shift, modes, menus & navigation`

Primary: Cursor ▶ — move right in the line

#13 OK `Operators & execute`

Primary: OK — confirm the highlighted item

#14 ON Clear, delete & power

Primary: ON — power on

#15 f(x)= Shift, modes, menus & navigation

Primary: Overlay curve

#16 HOME Shift, modes, menus & navigation

Primary: HOME — return to the main calculation screen

#17 MODE Shift, modes, menus & navigation

Primary: SETUP — angle unit, number format, display options

#18 SEL Functions

Primary: sel

#19 avg Functions

Primary: Mean of list k

shifted (2nd / f / SHIFT layer) (press #6 first): σ — Population s.d.

#20 s Functions

Primary: Sample s.d. of list k

#21 Med Functions

Primary: Median of list k

shifted (2nd / f / SHIFT layer) (press #6 first): Q1 — First quartile

#22 IQR Functions

Primary: Interquartile range

shifted (2nd / f / SHIFT layer) (press #6 first): Q3 — Third quartile

#23 n Functions

Primary: Count of list k

shifted (2nd / f / SHIFT layer) (press #6 first): out — Number of outliers

#24 P% Functions

Primary: Percentile

#25 bin# Functions

Primary: Count in histogram bin i

#26 b Functions

Primary: Regression slope

shifted (2nd / f / SHIFT layer) (press #6 first): a — Regression intercept

#27 **r** Functions

Primary: Correlation L1–L2

#28 **φ** Functions

Primary: Normal density

#29 **Φ** Functions

Primary: Normal probability a…b

shifted (2nd / f / SHIFT layer) (press #6 first): Φ^{-1} — Inverse normal

#30 **√** Functions

Primary: Square root √

shifted (2nd / f / SHIFT layer) (press #6 first): x^2 — Square x^2

Keyboard: r

#31 **(** Operators & execute

Primary: Opening bracket (

Keyboard: (

#32 **)** Operators & execute

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

Keyboard:)

#33 , Operators & execute

Primary: Comma / argument separator (between function arguments, list items or coordinates)

Keyboard: ,

#34 ^ Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

Keyboard: ^

#35 x! Functions

Primary: Factorial n! (also on non-integers via the gamma function where supported)

Keyboard: !

#36 π Functions

Primary: π (3.14159265…) as an exact constant

Keyboard: p

#37 7 Digits & number entry

Primary: Digit 7

Keyboard: 7

#38 8 `Digits & number entry`

Primary: Digit 8

Keyboard: `8`

#39 9 `Digits & number entry`

Primary: Digit 9

Keyboard: `9`

#40 DEL `Clear, delete & power`

Primary: DEL — delete the character at the cursor

Keyboard: `Backspace` `Delete`

#41 AC `Clear, delete & power`

Primary: AC — clear all of the current calculation

Keyboard: `Escape`

#42 4 `Digits & number entry`

Primary: Digit 4

Keyboard: `4`

#43 5 `Digits & number entry`

Primary: Digit 5

Keyboard: `5`

#44 6 Digits & number entry

Primary: Digit 6

Keyboard: 6

#45 × Operators & execute

Primary: Multiplication ×

Keyboard: * x

#46 ÷ Operators & execute

Primary: Division ÷

Keyboard: /

#47 1 Digits & number entry

Primary: Digit 1

Keyboard: 1

#48 2 Digits & number entry

Primary: Digit 2

Keyboard: 2

#49 3 Digits & number entry

Primary: Digit 3

Keyboard: 3

#50 **+** `Operators & execute`

Primary: Addition +

Keyboard: `+`

#51 **–** `Operators & execute`

Primary: Subtraction –

Keyboard: `-`

#52 **0** `Digits & number entry`

Primary: Digit 0

Keyboard: `0`

#53 **.** `Digits & number entry`

Primary: Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

Keyboard: `.`

#54 **(–)** `Digits & number entry`

Primary: Negative sign (–) — makes the next number negative. Different from the subtraction key

shifted (2nd / f / SHIFT layer) (press #6 first): Ans — Ans — recall the last answer into the current expression

#55 E `Digits & number entry`

Primary: ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23

Keyboard: `e`

#56 EXE `Operators & execute`

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Keyboard: `Enter` `=`

5 · Worked examples

7 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Mean of the sample exam scores L1

Calculate **avg(1)**. Press **Reset** first.

`#19` avg → `#47` 1 → `#32`) → `#56` EXE

Display shows `75.55` — expected 75.55. ✔ verified on the app.

Example 5.2 — Median of L1

Calculate **med(1)**. Press **Reset** first.

`#21` Med → `#47` 1 → `#32`) → `#56` EXE

Display shows `75` — expected 75. ✔ verified on the app.

Example 5.3 — Interquartile range of L1 (median-of-halves)

Calculate **iqr(1)**. Press **Reset** first.

`#22` IQR → `#47` 1 → `#32`) → `#56` EXE

Display shows `13.5` — expected 13.5. ✔ verified on the app.

Example 5.4 — Histogram bin 3 count (bins of width 10 from 50)

Calculate **bincount(3)**. Press **Reset** first.

#25 bin# → **#49** 3 → **#32**) → **#56** EXE

Display shows `8` — expected 8. ✔ verified on the app.

Example 5.5 — Least-squares slope of L1 on L2

Calculate **slope( )**. Press **Reset** first.

#26 b → **#32**) → **#56** EXE

Display shows `4.153447589` — expected 4.15344759. ✔ verified on the app.

Example 5.6 — Standard normal probability within one σ

Calculate **ncdf(−1,1)**. Press **Reset** first.

#29 Φ → **#54** (−) → **#47** 1 → **#33** , → **#47** 1 → **#32**) → **#56** EXE

Display shows `0.6826894921` — expected 0.68268949. ✔ verified on the app.

Example 5.7 — Correlation coefficient r of L1 and L2

Calculate **corrl( )**. Press **Reset** first.

#27 r → **#32**) → **#56** EXE

Display shows `0.989808376` — expected 0.98980838. ✔ verified on the app.

6 · Tricks & hacks

Hack 1: History reuse

Scroll up in the home screen history and press ENTER/EXE to copy an old entry or answer.

Hack 2: Ans chaining

Start an entry with an operator to continue from the previous answer.

Hack 3: Store variables

Use STO▶ / → to store values into letters A–Z, then use them in expressions.

Hack 4: Table of values

Use the TABLE view to scan where a function changes sign before solving.

Hack 5: Window first

If a graph looks empty, adjust the WINDOW / zoom standard.

Hack 6: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 7: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = ...

Hack 8: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 9: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 10: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 11: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 12: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 13: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 14: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 15: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 16: Hidden functions on this model

9 extra functions sit on shifted layers. Press the layer key once, then the key: for example #6 then #5 → TABLE; #6 then #19 → σ; #6 then #21 → Q1; #6 then #22 → Q3.

Hack 17: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left→right, top→bottom from #1 to #56.

Hack 18: Type faster on a computer

26 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 19: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 20: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 21: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
r	#30	√	Square root √
(	#31	(	Opening bracket (
)	#32	)	Closing bracket). Closing brackets at the end of a line may be omitted
,	#33	,	Comma / argument separator (between function arguments, list items or coordinates)
^	#34	^	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
!	#35	x!	Factorial n! (also on non-integers via the gamma function where supported)
p	#36	π	π (3.14159265…) as an exact constant
7	#37	7	Digit 7
8	#38	8	Digit 8
9	#39	9	Digit 9
Backspace Delete	#40	DEL	DEL — delete the character at the cursor
Escape	#41	AC	AC — clear all of the current calculation
4	#42	4	Digit 4
5	#43	5	Digit 5
6	#44	6	Digit 6
* x	#45	×	Multiplication ×
/	#46	÷	Division ÷
1	#47	1	Digit 1
2	#48	2	Digit 2

3	#49	3	Digit 3
+	#50	+	Addition +
-	#51	–	Subtraction –
0	#52	0	Digit 0
.	#53	.	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
e	#55	E	×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23
Enter =	#56	EXE	Execute / = — evaluate the line and show the result (EXE / ENTER / =)

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: (–) vs –

Use the negative key for negative numbers.

Mistake 2: Radian mode

Graphs of sin x in degrees look flat over −10..10; switch to radians for standard trig windows.

Mistake 3: Implicit multiplication

2/3x may be read as 2/(3x) depending on the model; use brackets.

Mistake 4: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 5: Wrong angle unit

If sin 30 gives −0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 6: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 7: Division precedence

1 ÷ 2π may be read as 1 ÷ (2π) on textbook models. Type 1 ÷ (2 × π) explicitly to be sure.

Mistake 8: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 9: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 10: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 11: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

Mistake 12: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

9 · Practice drills

Set A — Calculation drills (7)

Work each one on calculator #21, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. **avg(1)** = _____
2. **med(1)** = _____
3. **iqr(1)** = _____
4. **bincount(3)** = _____
5. **slope()** = _____
6. **ncdf(−1,1)** = _____
7. **corrl()** = _____

Set B — Key-finding drills (55)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “DATA” (Open or close the list editor)? _____
2. Which key is “PLOT” (Cycle plot type)? _____
3. Which key is “ZOOM” (ZoomStat and zoom menu)? _____
4. Which key is “TRACE” (TRACE)? _____
5. Which key is “GRAPH” (GRAPH)? _____
6. Which key is “SHIFT” (SHIFT / 2nd)? _____
7. Which key is “TOOLS” (Plot types, fits, summaries)? _____

8. Which key is “WINDOW” (Bins and view)? _____

9. Which key is “▲” (Cursor ▲)? _____

10. Which key is “▼” (Cursor ▼)? _____

11. Which key is “◀” (Cursor ◀)? _____

12. Which key is “▶” (Cursor ▶)? _____

13. Which key is “OK” (OK)? _____

14. Which key is “ON” (ON)? _____

15. Which key is “f(x)=” (Overlay curve)? _____

16. Which key is “HOME” (HOME)? _____

17. Which key is “MODE” (SETUP)? _____

18. Which key is “SEL” (sel)? _____

19. Which key is “avg” (Mean of list k)? _____

20. Which key is “s” (Sample s.d. of list k)? _____

21. Which key is “Med” (Median of list k)? _____

22. Which key is “IQR” (Interquartile range)? _____

23. Which key is “n” (Count of list k)? _____

24. Which key is “P%” (Percentile)? _____

25. Which key is “bin#” (Count in histogram bin i)? _____

26. Which key is “b” (Regression slope)? _____

27. Which key is “r” (Correlation L1–L2)? _____

28. **Which key is "φ" (Normal density)?** _____
29. **Which key is "Φ" (Normal probability a…b)?** _____
30. **Which key is "√" (Square root √)?** _____
31. **Which key is "(" (Opening bracket)?** _____
32. **Which key is ")" (Closing bracket). Closing brackets at the end of a line may be omitted)?** _____
33. **Which key is "," (Comma / argument separator)?** _____
34. **Which key is "^" (Power x^y / ^)?** _____
35. **Which key is "x!" (Factorial n!)?** _____
36. **Which key is "π" (π)?** _____
37. **Which key is "DEL" (DEL)?** _____
38. **Which key is "AC" (AC)?** _____
39. **Which key is "×" (Multiplication ×)?** _____
40. **Which key is "÷" (Division ÷)?** _____
41. **Which key is "+" (Addition +)?** _____
42. **Which key is "−" (Subtraction −)?** _____
43. **Which key is "." (Decimal point. Pressing it first gives "0."; a second point in the same number is ignored)?** _____
44. **Which key is "(−)" (Negative sign)?** _____
45. **Which key is "E" (×10^x / EXP / EE)?** _____
46. **Which key is "EXE" (Execute / =)?** _____

47. How do you reach "TABLE"? _____

48. How do you reach "σ"? _____

49. How do you reach "Q1"? _____

50. How do you reach "Q3"? _____

51. How do you reach "out"? _____

52. How do you reach "a"? _____

53. How do you reach "Φ^{-1}"? _____

54. How do you reach "x^2"? _____

55. How do you reach "Ans"? _____

Appendix A — Drill answers

Set A

1. **75.55** (display `75.55`)
   `#19` avg → `#47` 1 → `#32`) → `#56` EXE
2. **75** (display `75`)
   `#21` Med → `#47` 1 → `#32`) → `#56` EXE
3. **13.5** (display `13.5`)
   `#22` IQR → `#47` 1 → `#32`) → `#56` EXE
4. **8** (display `8`)
   `#25` bin# → `#49` 3 → `#32`) → `#56` EXE
5. **4.15344759** (display `4.153447589`)
   `#26` b → `#32`) → `#56` EXE
6. **0.68268949** (display `0.6826894921`)
   `#29` Φ → `#54` (−) → `#47` 1 → `#33` , → `#47` 1 → `#32`) → `#56` EXE
7. **0.98980838** (display `0.989808376`)
   `#27` r → `#32`) → `#56` EXE

Set B

1. #1
2. #2
3. #3
4. #4

5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #28
29. #29
30. #30
31. #31
32. #32
33. #33
34. #34
35. #35
36. #36
37. #40
38. #41
39. #45
40. #46
41. #50
42. #51
43. #53
44. #54
45. #55
46. #56
47. #6 SHIFT then #5 GRAPH
48. #6 SHIFT then #19 avg
49. #6 SHIFT then #21 Med
50. #6 SHIFT then #22 IQR
51. #6 SHIFT then #23 n
52. #6 SHIFT then #26 b
53. #6 SHIFT then #29 Φ
54. #6 SHIFT then #30 √
55. #6 SHIFT then #54 (-)

Appendix B — Function finder (A–Z)

Function	Label	Keys
×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23	E	#55
AC — clear all of the current calculation	AC	#41
Addition +	+	#50
Ans — recall the last answer into the current expression	Ans	#6 → #54
Bins and view	WINDOW	#8
Closing bracket). Closing brackets at the end of a line may be omitted	)	#32
Comma / argument separator (between function arguments, list items or coordinates)	,	#33
Correlation L1–L2	r	#27
Count in histogram bin i	bin#	#25
Count of list k	n	#23
Cursor ▲ — up / previous history line / previous item	▲	#9
Cursor ▶ — move right in the line	▶	#12
Cursor ▼ — down / next history line / next item	▼	#10
Cursor ◀ — move left in the line	◀	#11
Cycle plot type	PLOT	#2
Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored	.	#53
DEL — delete the character at the cursor	DEL	#40
Digit 0	0	#52
Digit 1	1	#47
Digit 2	2	#48
Digit 3	3	#49
Digit 4	4	#42
Digit 5	5	#43

Digit 6	**6**	**#44**
Digit 7	**7**	**#37**
Digit 8	**8**	**#38**
Digit 9	**9**	**#39**
Division ÷	**÷**	**#46**
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	**EXE**	**#56**
Factorial n! (also on non-integers via the gamma function where supported)	**x!**	**#35**
First quartile	**Q1**	**#6 → #21**
GRAPH — draw the graph	**GRAPH**	**#5**
HOME — return to the main calculation screen	**HOME**	**#16**
Interquartile range	**IQR**	**#22**
Inverse normal	Φ^{-1}	**#6 → #29**
Mean of list k	**avg**	**#19**
Median of list k	**Med**	**#21**
Multiplication ×	**×**	**#45**
Negative sign (−) — makes the next number negative. Different from the subtraction key	**(−)**	**#54**
Normal density	φ	**#28**
Normal probability a…b	Φ	**#29**
Number of outliers	**out**	**#6 → #23**
OK — confirm the highlighted item	**OK**	**#13**
ON — power on	**ON**	**#14**
Open or close the list editor	**DATA**	**#1**
Opening bracket (	**(**	**#31**
Overlay curve	**f(x)=**	**#15**
Percentile	**P%**	**#24**

Plot types, fits, summaries	TOOLS	#7
Population s.d.	σ	#6 → #19
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	^	#34
Regression intercept	a	#6 → #26
Regression slope	b	#26
Sample s.d. of list k	s	#20
sel	SEL	#18
SETUP — angle unit, number format, display options	MODE	#17
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	SHIFT	#6
Square root √	√	#30
Square x^2	x^2	#6 → #30
Subtraction −	−	#51
TABLE — table of values	TABLE	#6 → #5
Third quartile	Q3	#6 → #22
TRACE — move along a graph	TRACE	#4
ZoomStat and zoom menu	ZOOM	#3
π (3.14159265…) as an exact constant	π	#36

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

Statistics plotter. L1 is the main variable; scatter plots use L2 on the x-axis and L1 on the y-axis.

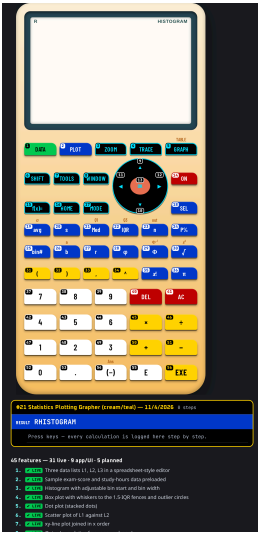
Quartiles use the median-of-halves rule; percentiles interpolate linearly between order statistics.

Planned (not yet working)

- **Graph image export (PNG)**
- **Shareable function list link**
- **Dark screen theme**
- **Speech read-out of values**
- **Exam mode**

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 DATA	#2 PLOT	#3 ZOOM
#4 TRACE	#5 GRAPH ⇧ TABLE	#6 SHIFT
#7 TOOLS	#8 WINDOW	#9 ▲
#10 ▼	#11 ◀	#12 ▶
#13 OK	#14 ON	#15 f(x)=
#16 HOME	#17 MODE	#18 SEL
#19 avg ⇧ σ	#20 s	#21 Med ⇧ Q1
#22 IQR ⇧ Q3	#23 n ⇧ out	#24 P%
#25 bin#	#26 b ⇧ a	#27 r
#28 φ	#29 Φ ⇧ Φ^{-1}	#30 √ ⇧ x^2
#31 (	#32)	#33 ,
#34 ^	#35 x!	#36 π
#37 7	#38 8	#39 9
#40 DEL	#41 AC	#42 4
#43 5	#44 6	#45 ×
#46 ÷	#47 1	#48 2
#49 3	#50 +	#51 –

#52 0	#53 .	#54 (−) ⇧ Ans
#55 E	#56 EXE	

U-45 CALCULATORS · Calculator #21 · cut out and keep